

**M.TECH.
(SEM-I) THEORY EXAMINATION 2020-23
ADVANCED ENGINEERING MATHEMATICS**

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****7 x 2 = 14**

- (a) Define norm of a vector.
- (b) Define basis of a vector space.
- (c) Differentiate between discrete and continuous random variables.
- (d) Differentiate between probability distribution function and probability density function.
- (e) Define power spectrum.
- (f) Define shot noise.
- (g) What is utilization factor of queueing model?

SECTION B**2. Attempt any three of the following:****3 x 7 = 21**

- (a) Let V be the vector space of all 2×2 real matrices. Show that the sets $S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right\}$ span V .
- (b) Suppose that $x(t)$ is a process with $\eta(t) = 3$ and $R(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$. Determine the mean and variance and then covariance of the random variables $z = x(5)$ and $w = x(8)$.
- (c) Let X and Y be jointly continuous random variables

$$f_{X,Y}(x, y) = \begin{cases} 6e^{-(2x+3y)} & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(X > Y)$

- (d) Explain Band limited process. If $x(t)$ is BL, then show that

$$E \left[\{x(t+\tau) - x(t)\}^2 \right] \leq \sigma^2 \tau^2 R(0)$$

OR equivalently

$$2[R(0) - R(\tau)] \leq \sigma^2 \tau^2 R(0)$$

- (e) Differentiate Random Process and Markov Process.

SECTION C

3. Attempt any *one* part of the following: 1 x 7 = 7
- (a) Let V be the vector space of all polynomials of degree ≤ 3 . Determine whether the set $S = \{t^3, t^2 + t, t^3 + t + 1\}$ spans V or not.
 - (b) Determine whether the set of vectors of the form (x, y, z) where $2x - y + z = 0$ forms a subspace of R^3 .
4. Attempt any *one* part of the following: 1 x 7 = 7
- (a) Let X be a continuous random variable and probability density function $f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \in R_X \\ 0 & x \notin R_X \end{cases}$ where $R_X = [0, \infty)$ find $E[e^{tX}]$
 - (b) Find the mean and variance of Poisson distribution.
5. Attempt any *one* part of the following: 1 x 7 = 7
- (a) State and prove Nyquist theorem.
 - (b) Let X and Y be jointly normal random variable with parameters $\mu_X = 1, \sigma_X^2 = 1, \mu_Y = 0, \sigma_Y^2 = 4$ and $Cov(X, Y) = \frac{1}{2}$. Find $P(2X + Y \leq 3)$
6. Attempt any *one* part of the following: 1 x 7 = 7
- (a) If $x(t)$ is a WSS process with autocorrelation $R(\tau) = Ae^{-\alpha|\tau|}$. Determine the second moment of the random variable $x(8) - x(5)$.
 - (b) Suppose $x(t) = a(t)\cos\omega_0 t - b(t)\sin\omega_0 t = r(t)\cos[\omega_0(t) + \phi(t)]$, where $a(t)$ and $b(t)$ with zero mean and a constant ω_0 and $r(t) = \sqrt{a^2(t) + b^2(t)}$ and $\tan\phi = \frac{b(t)}{a(t)}$. Show that $x(t)$ is WSS if the processes $a(t)$ and $b(t)$ are such that $R_{aa}(\tau) = R_{bb}(\tau)$ and $R_{ab}(\tau) = -R_{ba}(\tau)$
7. Attempt any *one* part of the following: 1 x 7 = 7
- (a) Differentiate between M/M/1 and M/G/1.
 - (b) A super market has two sales girls at the sales counters. If the service time for each customer is exponential with a mean of 4 minutes, and if the people arrive in a Poisson fashion at the rate of 10 an hour then calculate the
 - (i) Probability that a customer has to wait for being served?
 - (ii) Expected percentage of idle time for each sales girl?
 - (iii) If a customer has to wait, what is the expected length of his waiting time?